Presentation cheat sheet

Optimising CTU bus frequencies under demand variability

All demand is synthetic and illustrative. Say this at least twice.

1. The 2-minute elevator pitch

Say this

Chandigarh's CTU runs 68 bus routes with 389 buses. Some routes are badly overcrowded while others run nearly empty. We asked: **with the same fleet, can a better timetable serve people with less waiting and less crowding?**

We wrote the timetable as a mathematical optimisation problem. For every route and every part of the day (morning peak, midday, evening peak) the computer picks how often a bus runs, from a menu of headways such as 10, 15 or 20 minutes. The rules are hard: never more than 357 buses on the road, enough buses for each route's round trip, and service close to today's. We judge a timetable three ways: passenger waiting, kilometres driven, and passengers who do not fit on a bus.

Because those three goals conflict, we used goal programming to set targets and priorities, mapped every best compromise with two independent methods that confirm each other, and tested the result on 2000 random days of demand.

CTU does not publish ridership, so the demand is **synthetic**. In that setting the balanced plan cuts waiting by 3.3%, crowding overflow by about 30%, and kilometres by 3%, while using 357 buses instead of 385. The method is ready; real boarding data is the next step.

2. The 10-minute demo script

Time	Show	Say
0:00	Title slide or video scene 1	The problem: 68 routes, 389 buses, crowded and empty routes side by side. The question: same fleet, better service?
0:45	`methodology_block_diagram.png`	"Seven steps: check the data, calibrate, build demand, optimise, set goals, map trade-offs, simulate."
1:30	`calibration_N_need.png`	"Before optimising we checked our assumptions explain today's fleet. Reading route length as one-way, at 22 km/h with 10% layover, today's timetable needs 341 buses: 88% of 389. Reading it as a full trip leaves half the fleet idle, which is implausible."
2:30	`demand_heatmap.png`	"No ridership is published, so we built synthetic demand from route length and the big stops. Darker = busier. This is an assumption, not data."
3:15	`tfn_alpha_cuts.png`	"Demand is uncertain: we give a low, likely and high value, and plan capacity 17.5% above the likely one ($\alpha = 0.5$)."
4:00	Model slide / textbook Ch. 18	"The computer chooses one headway per route and period with yes/no switches. That keeps everything linear, so an exact solver (HiGHS) proves the best answer in under a second."
5:00	`gp_pct_change.png`	"Left: under today's frequency limits there is almost nothing to gain. Right: allowing slightly shorter headways, every priority order cuts unserved passengers 26–30% and waiting about 4%, on 357 buses."
6:15	`pareto_2d.png`	"These are all best compromises. Blue = exact method, orange = genetic algorithm. They lie on the same curve and the algorithm never beats an exact point: the two methods confirm each other. The star is today, far from the front."
7:15	`sim_overflow_distribution.png`	"On 2000 simulated days, today's timetable has more than 5% of passengers not fitting on 66% of days. The optimised plans: about 3%."
8:00	`sim_route_p95_LF.png`	"Where it comes from: buses move from routes with spare seats to the most crowded routes, like 31 (80 to 60 minutes)."
8:45	`sensitivity_tornado.png`	"Honesty: the result depends most on speed and layover. At 18 km/h no feasible timetable exists; at 26 km/h the gain doubles."

9:30	Limitations slide	"Synthetic demand, one speed, no depots or crews, CTU approval needed for shorter headways. Next step: real boarding data."

3. Each visual: what to point at, what to say

methodology_block_diagram Point down the boxes. "Each step feeds the next."

calibration_N_need Point at the black line (389) and the grey band, then the orange 22 km/h bar labelled 341. "Inside the band = the assumption fits today's fleet. Blue (S1) is far too low."

demand_heatmap Point at the dark top rows, then the lighter middle column. "Busiest routes at the top, midday lighter, grey = not running. Synthetic."

tfn_alpha_cuts Point at the triangle, then the orange dot at 0.5. "Low, likely, high. We plan for the 1.175 × likely value."

gp_pct_change Point at the empty left panel, then the tall green bars on the right. "Below zero is better. The rules on maximum frequency, not the fleet, were the bottleneck."

pareto_2d Point at the curve in the left panel, then the black star. "Less waiting costs more kilometres. Today is far up and right. Warning: 2-D shadows of a 3-D front."

pareto_3d Point at the thin sheet of points. "The front is a surface."

sim_tradeoff_bars Point at the blue bars left of zero in every panel. "Every optimised plan beats today on every measure."

sim_wait_vs_vkm Point at the black bubble top-right and the cluster bottom-middle. "Lower-left and smaller is better; note the wait axis spans only 0.6 minutes."

sim_overflow_distribution Point at the dashed 5% line and the black boxes to its right. "Today crosses 5% most days; optimised plans rarely."

sensitivity_tornado Point at the long top bar. "Speed matters most; the 22 km/h assumption needs CTU confirmation."

sensitivity_alpha Point at the flat blue lines. "Planning more cautiously barely changes things with a fixed fleet."

sensitivity_fleet Point at the single bar. "357 buses is near the minimum: 5% fewer and the minimum service cannot run."

sensitivity_out_of_sample Point at the orange bars. "On more uncertain demand everything gets worse, the optimised plans still much better than today."

sim_route_p95_LF Point at route 31 (grey 3.2 → blue 2.4) and route 32 (slightly worse). "Crowded routes improve; routes that gave up buses get a little fuller."

4. Likely questions with honest answers

1. **What problem are you solving?**

   Choosing how often buses run on each CTU route in three periods of the day, so that waiting and crowding fall without using more buses than exist.

2. **Is the data real?**

   The timetable is real (68 routes, 389 buses, 1662 km, 649 stops, as on 25.07.2026). The passenger demand is synthetic, because CTU does not publish ridership. Results show how the method behaves, not CTU outcomes.

3. **How did you create the demand?**

   A score per route from its length and whether it serves PGI, the ISBTs, outskirts or hub stops, times small random noise (fixed seed), scaled to about 81,000 daily boardings, then multiplied by 1.4 / 0.8 / 1.3 for AM / Midday / PM.

4. **Where does 81,326 come from?**

   143,000 CTU-wide daily riders scaled by 389/684 buses, from newspaper reports the project could not verify. It is flagged as unverified.

5. **What is a MILP?**

   A mixed-integer linear program: linear objective and constraints, some variables whole numbers or 0/1. Here: 0/1 headway switches, integer buses, continuous overflow.

6. **Why not just use waiting = 30/f?**

   It is a curve, and exact linear solvers need straight-line relations. Choosing the headway from a menu makes waiting a sum of fixed numbers times switches, which is linear and exact.

7. **What are the constraints?**

   At most 357 buses per period; enough buses for each route's cycle ($x \geq f\tau$); overflow at least demand minus capacity; frequency within allowed limits; buses close to today's numbers; no service where the route does not run; a soft cap on headways (60 min peak, 90 midday).

8. **Why 357 buses and not 389?**

   8% are kept as spares: $\lfloor 0.92 \times 389 \rfloor = 357$. With 10% spares (350) even the minimum published service is impossible (needs 352).

9. **Today uses 385 buses. How can you beat it with 357?**

   By moving buses from routes with spare seats (e.g. route 32: 14 → 11) to long, crowded, infrequent routes (31, 28A, 101: 80 → 60 min).

10. **What solver did you use?**

    HiGHS, an open-source solver, through the Python library PuLP. Each model solves in under a second, with proven optimality.

11. **How do you know the answers are correct?**

    Three checks: a 3-route toy where brute force over all 6,561 timetables gives exactly the solver's answer; every reported solution is re-checked against all constraints; and the genetic algorithm never beats an exact Pareto point.

12. **What is goal programming?**

    Giving each objective a target and minimising only the misses. Example: waiting at most 96% of today's. Priorities decide which target matters first.

13. **Which policy do you recommend?**

    P-BALANCED: equal weights on waiting, crowding, km and equity, because no evidence says one should strictly come first. It meets 3 of 4 targets (zero overflow is impossible by design).

14. **What is a Pareto front?**

    All timetables where no goal can improve without another getting worse. We found 46 exact points and 611 with the genetic algorithm.

15. **Why two methods (AUGMECON and NSGA-II)?**

    The exact method proves optimality; the genetic algorithm needs no linear formulas and could handle more realistic objectives later. Their agreement is a correctness check.

16. **NSGA-II has higher hypervolume. Isn't it better?**

    No. It has more points filling gaps between the grid points; it beats none of the exact points and stays close to them (GD ≈ 0.05).

17. **What is α?**

    How cautious capacity planning is. $\alpha = 1$ plans for the most likely demand, $\alpha = 0$ for the highest plausible. We use 0.5, i.e. 17.5% above most likely.

18. **Why does α barely matter?**

    With the fleet fixed, planning for more demand cannot add capacity; the optimiser can only shuffle the same buses.

19. **What does the simulation do?**

    2000 random days: each route's demand drawn from its low-likely-high triangle times a city-wide shock; gaps between buses irregular (CV 0.3). All timetables see the same days.

20. **Why does model waiting fall 4.1% but simulated waiting 3.3%?**

    The model uses most-likely demand and $h/2$; the simulation uses random demand, irregular buses and a per-passenger average.

21. **What matters most for the result?**

    Average speed and layover. At 26 km/h the waiting gain is 7.6%; at 18 km/h, or with 15% layover, no feasible timetable exists.

22. **Did you check reproducibility?**

    Yes. All random steps use fixed seeds; a fresh full run reproduces every saved number exactly (only run times differ).

23. **Why is overflow 2,249 passengers per day today?**

    That is in the synthetic demand: long infrequent routes like 31 have most-likely demand of 110 pax/h but capacity of only 45 pax/h.

24. **What is "relaxed $f_{\max}$"?**

    Allowing up to two menu levels shorter headways than the shortest published one (route 31: 80 → 45 or 60 min). Without it the model cannot help the crowded routes. CTU would need to approve it.

25. **Why is route length read as one way?**

    If it were the full trip, only 44% of the fleet would be needed and 57 of 68 routes would hold more buses than even their highest frequency needs. One-way gives 88%, which fits.

26. **What is the equity goal?**

    No headway above 60 minutes in the peaks or 90 at midday. Today exceeds it by 210 minutes in total; P-BALANCED by 0.

27. **Could CTU use this tomorrow?**

    Not directly: it needs real boarding counts, route-level speeds and depot and crew rules. The model and code are ready to take that data (see `docs/REAL_DEMAND_DATA.md`).

28. **What happens if demand is more uncertain than assumed?**

    Bad-day probability rises from 2.9% to 28.8% for P-BALANCED, and from 65.6% to 70.3% for today's timetable. Still better, but the gain depends on demand.

29. **Why did the "robust" variant do worse?**

    Planning every route for wider demand cannot add buses; it spreads the same capacity more thinly and weakens the weight on overflow. We report it as a negative result.

30. **How long does it run?**

    The whole pipeline about 4 minutes on a laptop; one MILP under a second; the 20 × 20 Pareto grid about 60 s.

5. What this project does NOT do (say it before they ask)

- It does not use real ridership; every result is synthetic and illustrative.
- It does not verify the 143,000-rider anchor.
- It does not model congestion or route-specific speeds (one speed, 22 km/h).
- It does not model transfers, route choice or demand reacting to better service.
- It does not model depots, driver shifts, crews or maintenance schedules.
- It does not optimise night and early-morning service (only 07:00–19:00).
- It does not model bunching's effect on capacity (only on waiting).
- It does not check terminal or stop capacity for the shorter headways.
- It does not change routes, only frequencies on existing routes.
- It does not give a money value to waiting or crowding.

6. Numbers to remember

routes / buses / km / stops	68 / 389 / 1662 / 649
usable fleet	357 (8% spares) vs 385 used today
active route-periods	191
model size	651 variables, 576 constraints (1,015 variables relaxed)
speed, layover, capacity	22 km/h, 10%, 60 pax/bus
P-BALANCED vs today (simulated)	wait -3.3%, overflow -29.5%, km -3.1%
bad days (overflow $> 5\%$)	65.6% today, 2.9% P-BALANCED
Pareto	46 exact points, 611 NSGA-II, 0 exact points beaten